

It contains:

- **INSTRUCTIONS** – background + sheet overview
- **Data** – cleaned 3 000-track Spotify dataset
- **Solutions - Correlation** – working CORREL formulas, strength/sign labels, full plot interpretations matching the official solution, and the three scatter plots with trend lines
- **Correlation Matrix** – colour-coded pairwise matrix of the main numeric features (so you can spot other interesting pairs)
- **Code Examples** – Python & R that reproduce every correlation and scatter plot

All formulas recalculated with zero errors; numbers match the official solution key exactly.

## Exercise 1 – Correlation (exact solutions)

| Features              | Correlation (r) | Strength    | Sig ↔ Options |
|-----------------------|-----------------|-------------|---------------|
| energy – loudness     | +0.7010         | Strong      | Positive      |
| energy – acousticness | −0.6109         | Moderate    | Negative      |
| energy – duration_ms  | +0.0174         | Weak / none | ≈ zero        |

Formulas used (Excel / Google Sheets):

excel

```
=CORREL(Data!G:G, Data!I:I) // energy - loudness
=CORREL(Data!G:G, Data!L:L) // energy - acousticness
=CORREL(Data!G:G, Data!D:D) // energy - duration_ms
```

Rule-of-thumb for  $|r|$ :

- $|r| \geq 0.7 \rightarrow$  Strong
- $0.3 \leq |r| < 0.7 \rightarrow$  Moderate
- $|r| < 0.3 \rightarrow$  Weak / negligible

## Plot-based answers (official solution language)

### 1. Energy vs Loudness

- Points fit a straight line with positive slope quite well  $\rightarrow$  **strong** correlation ( $|r| > 0.7$ ).
- Positive slope  $\rightarrow$  **positive** correlation. High-energy songs tend to be loud—intuitively sensible.

### 2. Energy vs Acousticness

- Points fit a straight line with negative slope only moderately  $\rightarrow$  **moderate** correlation (0.3–0.7).
- Negative slope  $\rightarrow$  **negative** correlation. More acoustic songs tend to have less energy (classic “unplugged” / MTV Acoustic sound).

### 3. Energy vs Duration

- Points do not follow a clear positive or negative trend; the trend line is essentially flat  $\rightarrow$  **very low / near-zero** correlation.

b. Because there is almost no linear relationship you cannot reliably tell the sign from the plot alone. High-energy songs exist at every duration.

**Extra note from the official solution**

You might have expected an even higher correlation for energy and loudness. The dispersed points at low energy values (quiet, low-energy tracks that still span a range of loudness) keep the coefficient from climbing higher.

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**Quick link back to Parts 1–2**

- Energy is left-skewed and relatively tight (Part 2) **and** strongly positively correlated with loudness → a high-energy playlist will almost automatically be loud.
- Energy's moderate negative correlation with acousticness aligns with the strong right skew of acousticness (most tracks are non-acoustic and energetic).
- Duration is essentially independent of energy, so you can freely mix short bangers and long anthems inside the same high-energy set.